PharmaTrust Distributors Ltd.

Project: Smart Contract Enabled Pharmaceutical Supply Chain System for Anti-Counterfeiting and Product Traceability
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1. Executive Summary

This Analysis Report presents the regulated revision of the Project Specification provided for “Smart Contract Enabled Pharmaceutical Supply Chain System for Anti-Counterfeiting and Product Traceability”. Following the awarding of the contract, conduct a formal analysis to assess the completeness, consistency, feasibility, and clarity of the stated requirements prior to progressing to the Software Requirements Specification (SRS) phase.

The analysis is conducted using the BCIC method (Breakdown, Clarify, Interpret, Categorize). The objective is to correct incongruities, resolve ambiguities, enhance requirements with vendor best practices, and maintain full consistency with the original PS propositions. During this process, the requirements were systematically reviewed to identify incongruities, ambiguities, and implicit assumptions using a team-defined Incongruous Requirements Checklist (IRC). Priority issues were clarified with stakeholders, corrected through controlled rewriting, and enhanced with vendor best practices while maintaining consistency with the original intent of the Project Specification. A regulated-revision approach is adopted to minimise downstream risks during design, development, and acceptance.

Given the regulatory-critical nature of pharmaceutical supply chains (HSA, MOH, PDPC), requirement ambiguity presents elevated operational and compliance risk. Prior to analysis, 10% of business requirements contained unverifiable terms or unbounded scope. Following BCIC-based regulated revision, all critical requirements were converted into measurable, testable, and auditable statements suitable for contractual and regulatory validation. As a result, the vendor can proceed to SRS specification with a stable, regulator-ready requirements baseline and reduced delivery and acceptance risk.

2. BCIC – Analysing Activities

IRC Used:
https://docs.google.com/document/d/1fzCrPq0fRDCJ7avzG1QiBMSKLRobR_dT7ypfkDOri-E/edit?usp=sharing

2.1 Breakdown

The analysis focused specifically on identifying incongruent requirements within the Project Specification, as these represent the primary sources of delivery, acceptance, and compliance risk from a vendor perspective. The Breakdown activity was conducted using a structured Incongruous Requirements Checklist (IRC) as the analytical instrument, under which each business and technical requirement was evaluated against criteria including ambiguity, testability, consistency, feasibility, and completeness.

In addition, requirements were assessed for alignment with industry-standard external systems and regulatory frameworks, as well as for whether responsibilities were expressed as atomic, single-purpose obligations or improperly compounded within a single statement. Requirements that failed one or more IRC checks—due to omissions in external system scope, undefined validation authority or evidence mechanisms, infeasible assumptions, or conflation of distinct responsibilities such as authentication and MFA enforcement, order processing and verification, or backup and recovery objectives were classified as incongruent and documented in the breakdown tables.

Requirements that satisfied all IRC criteria were deemed congruent and intentionally excluded to maintain analytical focus on risk-relevant deficiencies. By isolating and documenting only these incongruent cases, the Breakdown establishes a clear, systematic, and traceable problem set that highlights where the original specification lacks precision or enforceability, thereby providing a focused foundation for subsequent clarification and interpretation activities.

Business Requirements

BR ID	Requirement Summary	Congruent / Incongruent	Compound / Atomic	External System / Industry Check	Issue Identified
BR-01	Secure password registration	Incongruent	Atomic	OWASP ASVS, NIST SP 800-63	“Secure” is undefined and not objectively verifiable.
BR-02	Login with MFA	Incongruent	Compound	ISO 27001, OWASP ASVS	Authentication and MFA enforcement are conflated.
BR-03	Operate blockchain network	Incongruent	Atomic	Blockchain governance practices	Integrity monitoring and evidence mechanisms not defined.
BR-04	Smart contracts created manually	Incongruent	Compound	Stratis smart contract lifecycle	Contract creation, deployment, and assignment not separated.
BR-05	Supplier order management	Incongruent	Compound	ERP / SCM systems	Order handling and verification responsibilities are unclear.
BR-06	Manufacturer medicine production	Incongruent	Atomic	GMP (HSA, WHO)	Regulatory compliance assumed but not enforced.
BR-07	Wholesaler distribution	Incongruent	Atomic	GDP compliance systems	Compliance stated without audit or evidence mechanism.
BR-08	Distributor order fulfilment	Incongruent	Atomic	Supply-chain verification platforms	Authenticity verification mechanism unspecified.
BR-09	Customer medicine ordering	Incongruent	Compound	Epic EHR / e-Prescription systems	Ordering and prescription validation not separated.
BR-10	Medication source verification	Incongruent	Atomic	DSCSA / traceability standards	Provenance retrieval scope undefined.
BR-11	Transporter logistics management	Incongruent	Atomic	GPS / fleet-tracking platforms	“Real-time” tracking is infeasible and unbounded.
BR-13	Custody transfer recording	Incongruent	Atomic	Blockchain custody systems	Timestamp and signature requirements unclear.
BR-14	Order status tracking	Incongruent	Atomic	ERP / SCM platforms	“Real-time” update expectation undefined.
BR-15	Product master data management	Incongruent	Atomic	HSA product registries	Validation authority not specified.

BR-16	Regulatory compliance reporting	Incongruent	Atomic	MOH, HSA, PDPC systems	Report outputs and scope undefined.
BR-17	Epic EHR integration	Incongruent	Compound	HL7 / FHIR ecosystems	Integration scope unbounded.
BR-18	SAP ERP integration	Incongruent	Atomic	Enterprise ERP systems	Synchronisation frequency unspecified.
BR-19	Tamper detection and alerts	Incongruent	Atomic	SOC / SIEM platforms	Alert timing requirements undefined.

Technical Requirements

TR ID	Requirement Summary	Congruent / Incongruent	Compound / Atomic	External System / Industry Check	Issue Identified
TR-11	CLI tools for admin operations	Incongruent	Compound	DevOps tooling standards	“Advanced operations” is unbounded and ambiguous.
TR-18	API backward compatibility	Incongruent	Atomic	API lifecycle management standards	Versioning and deprecation rules not defined.
TR-27	Backup & recovery	Incongruent	Compound	ISO 27001 / DR standards	Backup and recovery objectives conflated.
TR-32	Data migration tools	Incongruent	Atomic	ETL / data governance practices	Migration validation criteria missing.

2.2 Clarify

The Clarification activity focused on determining the priority incongruent requirements whose ambiguity or incompleteness could not be resolved through textual discussion alone, by directly cooperating with the originating stakeholders through zoom sessions.The objective of clarification was not to redesign requirements, but to uncover the authentic intent, constraints, and acceptance expectations held by the originating stakeholders. During these sessions, specific difficulties associated with the requirements were explained using user-friendly and **informative models** to support shared understanding. Hence reaching an agreement between the stakeholders on the requirements.

Req ID	Key Difficulty Identified	Model Needed	Purpose of Model
BR-01 (Password security)	Meaning of “secure” unclear	Data Dictionary	Used to precisely define password attributes (e.g. length, complexity, hashing) that cannot be unambiguously expressed through narrative text.
BR-07 (GDP compliance)	Compliance stated but unverifiable	Context Diagram	Used to visualise system boundaries and interactions with wholesalers and audit authorities, clarifying where and how compliance is verified.

BR-14 (Real-time order status)	“Real-time” undefined	Sequence / Dialog Map	Used to reason about event ordering and timing expectations, which cannot be validated using static descriptions.
BR-17 (Epic EHR integration)	Integration scope unbounded	Context Diagram	Used to constrain external data exchanges by clearly showing permitted interactions with Epic EHR and excluding non-essential flows.

BR-01 Data Dictionary

Element	Type	Constraints / Format	Description
password_plaintext	String	Never stored	User-entered password
password_min_length	Integer	e.g. 12+	Minimum password length requirement
password_complexity_rules	String / Enum	e.g. upper/lower/digit/symbol	Password composition policy
password_hash_algo	String	e.g. bcrypt/Argon2id	Hash algorithm for password
password_hash	String	Output of hash algo	Stored password hash
password_salt	String	Auto-generated	Salt applied by hashing scheme
password_failed_attempt_count	Integer	0..N	Count of consecutive failed attempts
account_lockout_threshold	Integer	e.g. 5 attempts	Max failures before lock
account_lockout_duration	Duration	e.g. 15 minutes	Lockout time window

2.3 Interpret

Each incongruent requirement was fully revised in its entirety, rather than partially amended, to eliminate ambiguity, inappropriate imperative usage, feasibility concerns, and incomplete scope. Revised requirements were expressed using standardised imperatives (e.g. *shall*) and measurable conditions, ensuring alignment with IEEE-style SRS conventions and improving verifiability.

In addition, industry-standard codification was applied to revised requirements to replace informal or customer-specific phrasing embedded in the original Project Specification. This ensured that all interpreted requirements were expressed in a neutral implementation-independent form, appropriate for contractual and engineering use.

The revised business and technical requirement statements documented in the analysis tables below, which now represent a complete and congruent set of requirements ready for categorisation and inclusion in the SRS.

BR ID	Incongruity Identified (from Breakdown)	Cogent Business Requirement (Interpret Output)
BR-01	“Secure” password undefined	The system shall enforce password policies including minimum length, complexity rules, and secure cryptographic hashing for all user accounts.

BR-02	Mixed obligation wording	The system shall enforce multi-factor authentication for all administrative accounts and shall authenticate users using verified credentials.
BR-03	Integrity monitoring not defined	The system shall operate a permissioned blockchain network and shall monitor ledger integrity to detect unauthorised modification events.
BR-04	Compound smart-contract lifecycle	Administrators shall create, deploy, and assign immutable smart contracts using the Stratis platform.
BR-05	Weak imperative for core function	Suppliers shall manage raw-material orders and shall update order status with blockchain verification.
BR-06	GMP compliance implicit	Manufacturers shall manage medicine production in compliance with verified GMP certification.
BR-07	GDP compliance unverifiable	Wholesalers shall process and verify orders in compliance with GDP requirements, with audit logs recorded for regulatory review.
BR-08	Authenticity verification unclear	Distributors shall verify product authenticity using blockchain custody records prior to order fulfilment.
BR-09	Prescription validation missing	Customers shall place medicine orders only after successful prescription validation, where applicable.
BR-10	Vague provenance verification	The system shall enable users to verify medicine provenance by retrieving complete custody history from the blockchain.
BR-11	“Real-time” GPS unrealistic	Transporters shall provide periodic GPS location updates during shipment transit.
BR-13	Timestamp requirements unclear	Each custody transfer shall be recorded as a blockchain transaction with UTC timestamp and digital signatures.
BR-14	“Real-time” order status ambiguous	Order status shall be updated within five seconds of blockchain transaction confirmation.
BR-15	Validation authority missing	Product master data shall validate NDC codes against the HSA registry prior to approval.
BR-16	Compliance reporting undefined	The system shall generate regulatory compliance reports aligned with MOH, HSA, and PDPC requirements.
BR-17	Integration scope unbounded	The system shall integrate with Epic EHR using HL7/FHIR APIs solely for prescription validation.
BR-18	ERP synchronisation frequency unclear	The system shall synchronise financial and inventory data with SAP ERP on a daily basis.
BR-19	Alert timing undefined	The system shall issue tamper alerts within 30 seconds of detection.

TR ID	**Incongruity Identified (from Breakdown)**	**Cogent Technical Requirement (Interpret Output)**
TR-11	“Advanced admin operations” ambiguous	The system shall provide CLI tools limited to deployment, configuration, and monitoring functions, as defined in the administrator operations specification.
TR-18	Backward compatibility incomplete	The system shall maintain backward compatibility for the two most recent API versions and shall publish a documented deprecation policy.

TR-27	Backup & recovery not testable	The system shall perform daily automated backups with a Recovery Point Objective (RPO) of 24 hours and a Recovery Time Objective (RTO) of 4 hours.
TR-32	Migration validation unclear	The system shall provide data-migration tools with post-migration validation reports confirming record completeness and integrity.

3. Repository

A centralised requirements repository was established as the single source of truth for all analysed requirements, supporting artefacts, and clarification models produced during the analysis phase. The repository provides structured governance over requirement evolution by ensuring consistency, traceability, and controlled access across analysis, clarification, and subsequent specification activities. It enables traceability from original project specifications clauses through Breakdown, Clarify models, and interpreted requirements for the vendor team to manage changes systematically, avoid duplication or divergence of requirements, and maintain a stable baseline suitable for transition into the Software Requirements Specification (SRS).

The repository is publicly accessible for transparency and review at:
https://github.com/Ryan-wong123/ICT2113-Pharma-Supply-Chain

The repository is organised to reflect the requirements-engineering lifecycle and to clearly separate artefacts by purpose:

- **/requirements**
  Stores the analysed and refined requirements, including breakdown classifications and traceability information, serving as the authoritative reference for requirement intent and status.

- **/models**
  Contains all clarification models (e.g. context diagrams, use case diagrams, sequence/dialogue maps, and data dictionaries) used to resolve priority incongruities and support stakeholder validation.

4. Account Manager

From a vendor governance perspective, the account manager functioned as the primary risk owner and boundary manager throughout the regulated-revision process. By converting subjective expectations (e.g. “real-time” performance and compliance assumptions) into measurable, evidence-backed definitions, the account manager prevented scope creep, acceptance disputes, and unrealistic performance commitments that could undermine delivery and regulatory compliance. The structured facilitation of zoom sessions, clarification of decision-making authority, and use of demonstrable technical evidence ensured that requirement changes were controlled, traceable, and formally agreed upon. This approach not only stabilised the requirements baseline but also safeguarded vendor credibility, reduced downstream rework risk, and established a transparent, trust-based working relationship between the vendor and customer organisations.

The account manager from the customer side helped with facilitating the regulated-revision process :

- BR-11 and BR-14 both had specified “real-time” updates but customer stakeholders had different expectations (some expected sub-second responses, others interpreted as “current”). The account manager organised a joint workshop with customer IT staff and operations manager to establish a measurable definition of “real-time”. The account manager facilitated the discussion and negotiated consensus on “within 5-seconds” for order status (BR-14) and “periodic updates” on GPS tracking (BR-11).

- Multiple customer stakeholders provided conflicting guidance on GDP compliance requirements based on BR-07. The account manager helped to identify and document appropriate decision-making authority within the customer organisation. Also helping to establish a clear escalation path and obtaining formal sign-off from authorised representations, preventing requirement churn.

- During clarification sessions for blockchain and smart contract requirements (BR-03, BR-04, TR-21), customer stakeholders struggled to understand some technical concepts such as “immutable smart contracts”. This resulted in inability to validate whether proposed technical solutions met their business needs and delayed decision-making on critical technical requirements. The account manager created simplified visual aids by developing layman-friendly diagrams showing how smart contracts automates verification and how tamper detection works. The account manager also brought in Medicare Plus’s IT administrators to bridge the gap and translate technical concepts into business language. Customer stakeholders gained sufficient understanding to make informed decisions on technical requirements. It also helped to reduce risk of misunderstandings during later project phases and increased confidence leading to stronger vendor-customer relationship.

- Customer expectation of sub-second blockchain confirmation (TR-44) conflicted with the technical reality of permissioned blockchain networks. The account manager helped to arrange a technical demonstration of Stratis blockchain performance. This allowed the negotiation of acceptable performance thresholds with supporting evidence. The customer accepted 3-5 second confirmation time, which aligned with technical capabilities.

Risks Analysis
These risks are directly derived from the incongruities identified during Breakdown and are mitigated through clarified and interpreted requirements recorded in the repository

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
VR-01	**Requirement Volatility** - Customer frequently changes or expands requirements after development begins (scope creep)	High	High	Formal change request process, requirement freeze per sprint, impact analysis before approval
VR-02	**Ambiguous Requirements** - Business users provide high-level or unclear expectations leading to misinterpretation	Medium	High	BCIC/IRC analysis, structured elicitation workshops, prototypes for early validation
VR-03	**Slow Stakeholder Approvals** - Multiple departments (MOH, HSA, compliance teams) delay sign-offs and decisions	High	High	Pre-scheduled approval cycles, escalation path via account manager
VR-04	**Conflicting Stakeholder Priorities** - Compliance, IT, and operations teams have competing objectives	Medium	Medium	Joint workshops and prioritisation matrix (P1–P3 alignment)
VR-05	**Limited User Engagement** - Healthcare providers or logistics staff may resist participation in testing or workshops	Medium	Medium	Early involvement, demonstrations, training sessions, change champions

VR-06	**Integration Dependency Delays** - Customer-controlled systems (SAP/EHR) are unavailable or poorly documented	Medium	High	Early technical discovery, sandbox environments, mock APIs
VR-07	**Data Access Restrictions** - Security or PDPA policies prevent vendor from accessing necessary data for testing	Medium	High	Data anonymisation, secure test datasets, formal data-sharing agreements
VR-08	**Expectation Misalignment** - Customer expects features beyond agreed contract scope	Medium	High	Clear SRS baseline, signed acceptance criteria, scope management
VR-09	**Regulatory Changes Mid-Project** - New healthcare regulations require redesign or rework	Low	High	Modular architecture, compliance monitoring, contingency buffer
VR-10	**Knowledge Dependency Risk** - Key customer personnel unavailable or leave project	Medium	Medium	Documentation, cross-training, multiple points of contact

* Risks are continuously monitored and updated in the repository risk register.

5. Sign-Off

This Analysing Report confirms that the Project Specification has undergone regulated-revision and is ready for transition into the Software Requirements Specification phase.

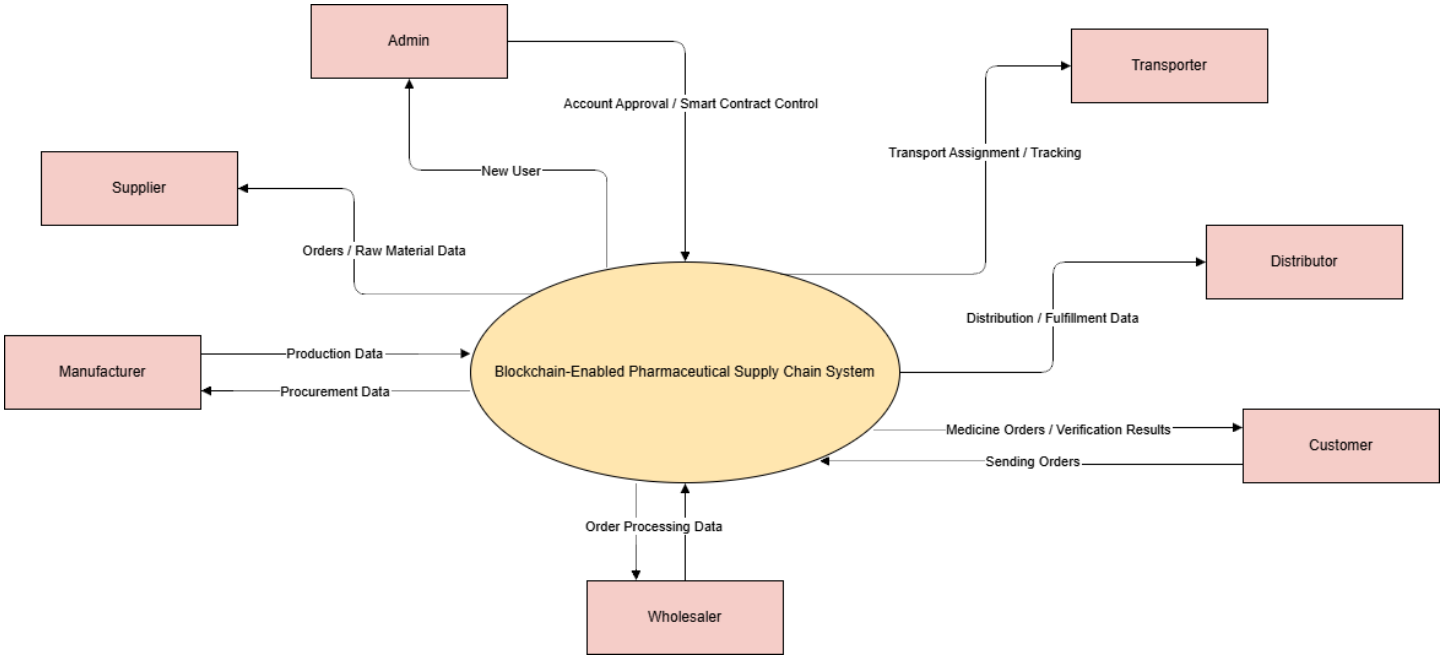
Signed:
Lead Requirements Analyst (Vendor)
Signature: _____
Date: _____

Status: Ready for submission to customer C-level stakeholders and commencement of Specifying phase.

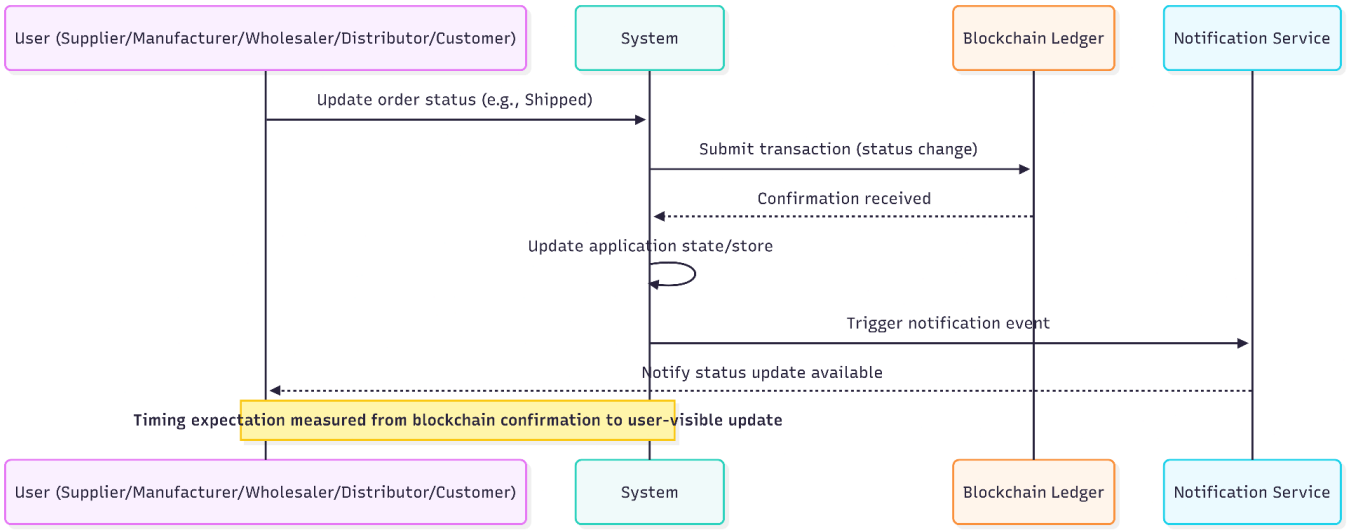
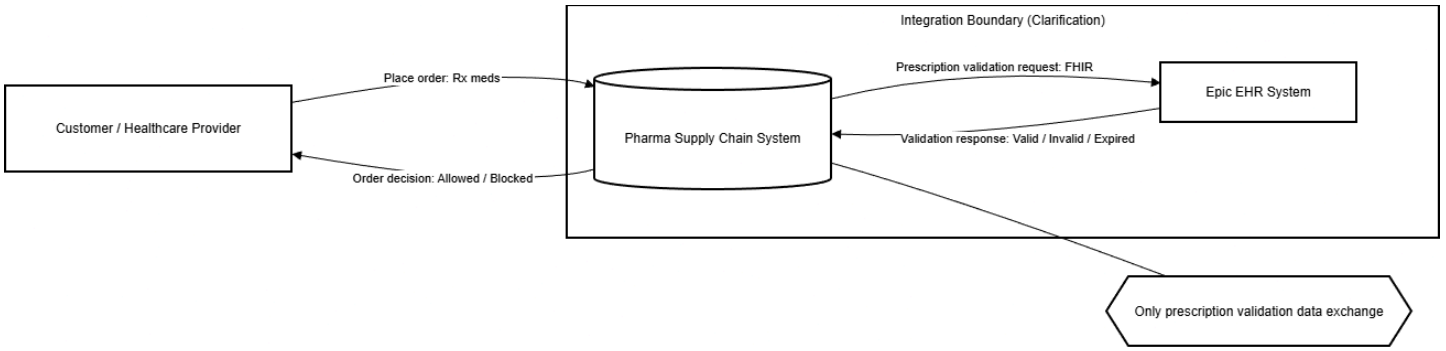
--END OF ANALYSIS REPORT--

Appendix A - Diagrams

BR-07 Context Diagram



BR-14 Sequence Diagram



BR-17 Context Diagram